

Aplasia of Lacrimal and Salivary Glands (ALSG): A Comprehensive Disease Characterization

Disease: Aplasia of Lacrimal and Salivary Glands **MONDO ID:** MONDO:0008397 · **OMIM:** #180920 · **Category:** Mendelian (autosomal dominant) **Causal gene:** *FGF10* (HGNC:3853; 5p12) **Evidence basis:** Literature synthesis (human clinical case reports/kindreds, ex vivo organ culture, mouse developmental genetics). Aggregated disease-level synthesis; no patient-level/EHR data.

Summary

Aplasia of Lacrimal and Salivary Glands (ALSG) is a rare, congenital, autosomal dominant Mendelian disorder caused by heterozygous loss-of-function variants in the fibroblast growth factor 10 gene (*FGF10*), producing **haploinsufficiency**. The disorder is characterized by aplasia, atresia, or hypoplasia of the lacrimal and salivary gland systems, leading to lifelong dry eyes (xerophthalmia), excessive tearing from blocked drainage (epiphora), and dry mouth (xerostomia). ALSG sits at the mild end of a phenotypic spectrum that is allelic with the more severe **Lacrimo-Auriculo-Dento-Digital (LADD) syndrome**, which is caused by mutations in *FGF10* and in its receptors *FGFR2* and *FGFR3*. Whereas ALSG is restricted to lacrimal and salivary involvement, LADD adds ear, dental, and digital anomalies.

Mechanistically, ALSG is a **dosage-sensitive developmental signaling disorder**. During embryogenesis, FGF10 secreted from gland mesenchyme activates epithelial FGFR2b (FGFR2-IIIb) and downstream ERK1/2 signaling to drive the branching morphogenesis and duct elongation that build the lacrimal and salivary glands. Reduced FGF10 gene dosage fails to adequately drive this epithelial program, so the glands never form properly. This mechanism is firmly supported by mouse genetics: *Fgf10*-null mice fail to develop lacrimal and salivary glands, and blocking FGFR2-IIIb suppresses endogenous lacrimal bud development. The clinical consequences are secondary and largely preventable: chronic xerostomia produces

rampant dental caries (approaching 100% of reported agenesis cases), oral candidiasis, and ascending sialadenitis, while alacrimia produces recurrent corneal erosions, ulceration, scarring, and sight-threatening keratopathy.

ALSG is ultra-rare, reported only in scattered multigenerational kindreds and sporadic cases, with no validated population prevalence. It shows **incomplete penetrance and marked intrafamilial variable expressivity**, and each child of an affected parent has a 50% risk of inheriting the variant. Lifespan is normal, and there is **no disease-modifying therapy**. Management is entirely supportive: aggressive caries prevention with fluoride and saliva substitutes, artificial tears, lacrimal drainage reconstruction, and, for severe congenital alacrimia with corneal damage, autologous minor salivary gland transplantation (shown to improve corneal transparency and neovascularization at 5-year follow-up). This report synthesizes 11 confirmed findings across 21 reviewed papers into a complete disease knowledge base entry spanning all 15 template sections.

1. Disease Information

Overview. ALSG is a congenital Mendelian disorder defined by aplasia, atresia, or hypoplasia of the lacrimal and salivary glands and ducts. Affected individuals present with a combination of dry eyes, excessive tearing (epiphora due to nasolacrimal drainage obstruction or punctal atresia), and dry mouth. As one primary source states: *"Aplasia of lacrimal and salivary glands (ALSG) is a rare autosomal dominant inherited disease, characterized by aplasia, atresia, or hypoplasia of the lacrimal and salivary systems with variable expressivity"* (PMID: 26955834).

Key identifiers:

Resource	Identifier
MONDO	MONDO:0008397
OMIM	#180920 (Aplasia of lacrimal and salivary glands; ALSG)
Gene	<i>FGF10</i> (HGNC:3853), chromosome 5p12
Allelic disorder	LADD syndrome (OMIM #149730)
Inheritance	Autosomal dominant

Synonyms / alternative names: ALSG; Aplasia of lacrimal and salivary glands; Aplasia/hypoplasia of lacrimal and salivary glands; Lacrimal and salivary gland aplasia. Related/overlapping entity: Lacrimo-auriculo-dento-digital syndrome (LADD; Levy-Hollister syndrome).

Information source type. This report is derived from **aggregated disease-level resources** (OMIM, published case reports, kindred studies, and systematic reviews) rather than individual EHR data. The evidence base consists predominantly of multigenerational family studies and single-case reports, consistent with an ultra-rare Mendelian disorder.

2. Etiology

Primary cause — genetic. ALSG is a monogenic disorder caused by **heterozygous loss-of-function variants in *FGF10*** that reduce the amount of functional FGF10 ligand by half (haploinsufficiency). *"This report further confirms that ALSG is caused by the haploinsufficiency of functional *FGF10*"* (PMID: 26955834).

Genetic risk factors. The disease is causally determined by the *FGF10* variant itself; there are no established susceptibility loci or GWAS signals (consistent with Mendelian etiology). A candidate second-locus/modifier gene, *IGSF3*, has been reported in the context of bony congenital nasolacrimal duct obstruction (see Section 4), but its role as a modifier remains to be established (PMID: 41533940).

Environmental risk factors. None are established. ALSG is a congenital developmental disorder with no known environmental, toxic, infectious, or lifestyle contribution to causation.

Protective factors. No genetic or environmental protective factors have been identified. Because the glands fail to form during embryogenesis, there is no modifiable exposure window after birth that alters disease occurrence (only secondary complications are modifiable).

Gene–environment interactions. None demonstrated. The phenotype is determined by *FGF10* dosage; however, the observed **incomplete penetrance and variable expressivity** (Section 4, Finding F004) imply the existence of unidentified genetic modifiers and/or stochastic developmental factors influencing how a given variant manifests.

3. Phenotypes

ALSG phenotypes fall into two anatomical domains — ocular/lacrimal and oral/salivary — plus rare extended features. All are **congenital** in onset (present from birth), though clinical recognition often occurs in early childhood (mean age ~6.4 years in a salivary agenesis review; [PMID: 41791508](#)). Severity is **variable**; progression of the primary gland defect is **stable** (non-progressive aplasia), but secondary complications (caries, keratopathy) are **progressive** if untreated.

Phenotype	Type	Suggested HPO term	Frequency / notes
Xerostomia (dry mouth)	Symptom	HP:0000217	80% of major salivary agenesis cases (PMID: 41791508)
Dental caries (rampant)	Clinical sign	HP:0000670	100% of reviewed agenesis cases (PMID: 41791508)
Aplasia/hypoplasia of salivary glands	Physical manifestation	HP:0009746 (abnormal salivary gland morphology)	Parotid, submandibular, sublingual
Decreased lacrimation / alacrimia	Symptom	HP:0000633 (xerophthalmia); HP:0000522 (alacrima)	~50% lacrimal involvement (PMID: 41791508)
Epiphora (excessive tearing)	Symptom	HP:0009926	Unilateral or bilateral; from drainage obstruction (PMID: 37615310)
Aplasia/atresia of lacrimal gland	Physical manifestation	HP:0000562 (abnormal lacrimal duct morphology)	Variable
Lacrimal punctal atresia/anomaly	Clinical sign	HP:0000562	Reported with gland agenesis (PMID: 22339588)
Recurrent corneal erosion / keratopathy	Clinical sign	HP:0000495	Sight-threatening in severe alacrimia (PMID: 38537898)
Oral candidiasis	Clinical sign	HP:0100580	Secondary to xerostomia (PMID: 22339588)
Bony congenital nasolacrimal duct obstruction	Physical manifestation	HP:0000579 (abnormal nasolacrimal system)	Novel extended phenotype (PMID: 38081329)

Quality-of-life impact. Chronic xerostomia impairs eating, swallowing, speech, and taste, and promotes relentless dental disease requiring continuous intervention. Alacrimia causes photophobia, ocular pain from corneal erosions, and risk of vision loss. Minor salivary gland

transplantation for severe alacrimia produced "*considerable improvement in the photophobia, corneal transparency, and neovascularization*" (PMID: 38537898), underscoring both the baseline burden and the modifiability of ocular morbidity.

4. Genetic / Molecular Information

Causal gene. *FGF10* (fibroblast growth factor 10), HGNC:3853, chromosome 5p12; encodes a paracrine FGF ligand that signals through FGFR2b.

Pathogenic variant spectrum. ALSG-causing variants are heterozygous and loss-of-function, acting through haploinsufficiency. Documented variant classes:

Variant	Type	Mechanism	Reference
c.237G>A (p.Trp79*) in exon 1	Nonsense	NMD-mediated haploinsufficiency	PMID: 26955834
c.429+1G>T (donor splice site)	Splice-site	Exon 2 skipping	PMID: 37615310
c.429+2T>A (donor splice site)	Splice-site	Aberrant splicing	PMID: 41533940
6 missense variants	Missense	Reduced ligand function	PMID: 41533940
5p12 deletion encompassing <i>FGF10</i>	Structural (CNV)	Whole-gene loss	PMID: 41533940
Novel LoF variant (EA/ALSG/PH case)	Loss-of-function	Haploinsufficiency	PMID: 41409311

Variant classification (ACMG/AMP). Nonsense, canonical splice-site, and whole-gene deletion variants meet pathogenic criteria (PVS1 null-variant in a LoF-mechanism gene plus segregation). Missense variants require case-by-case classification.

Allele frequency. Pathogenic variants are private/ultra-rare and essentially absent from population databases (gnomAD), consistent with a highly penetrant Mendelian disorder.

Somatic vs germline. All reported ALSG variants are **germline**.

Functional consequence. Loss of function → haploinsufficiency. A 50% reduction in functional FGF10 ligand is insufficient to drive normal gland morphogenesis, indicating the developing lacrimal/salivary epithelium is exquisitely dosage-sensitive to FGF10.

Allelism with LADD. ALSG and LADD form a single FGF-signaling disease spectrum. *"We identified heterozygous mutations in the tyrosine kinase domains of the genes encoding fibroblast growth factor receptors 2 and 3 (FGFR2, FGFR3) in LADD families, and in one further LADD family, we detected a mutation in the gene encoding fibroblast growth factor 10 (FGF10), a known FGFR ligand" (PMID: 16501574). "Lacrimo-auriculo-dento-digital (LADD) syndrome is a rare autosomal dominant disorder caused by mutations in one of the three genes: fibroblast growth factor receptor 2 (FGFR2), FGFR3, or FGF10" (PMID: 32715658).*

Modifier genes / second locus. *IGSF3* variants (three reported) co-occurring with *FGF10* variants in bony congenital nasolacrimal duct obstruction are candidate modifiers or an independent contributing locus (PMID: 41533940).

Epigenetic / chromosomal. No disease-specific epigenetic signature is established. The principal structural abnormality is the 5p12 microdeletion encompassing *FGF10*, detectable by chromosomal microarray (PMID: 41533940).

5. Environmental Information

No environmental, lifestyle, or infectious etiological factors are relevant to ALSG causation — it is a purely genetic developmental disorder. Environmental factors are relevant only to **secondary complication management**: dietary sugar intake and oral hygiene modulate the rate of caries progression, and ambient dryness/wind exacerbate ocular surface disease. Infectious agents (e.g., *Candida albicans* causing oral candidiasis; ascending bacterial sialadenitis) are **downstream consequences** of glandular hypofunction, not causes of the disease.

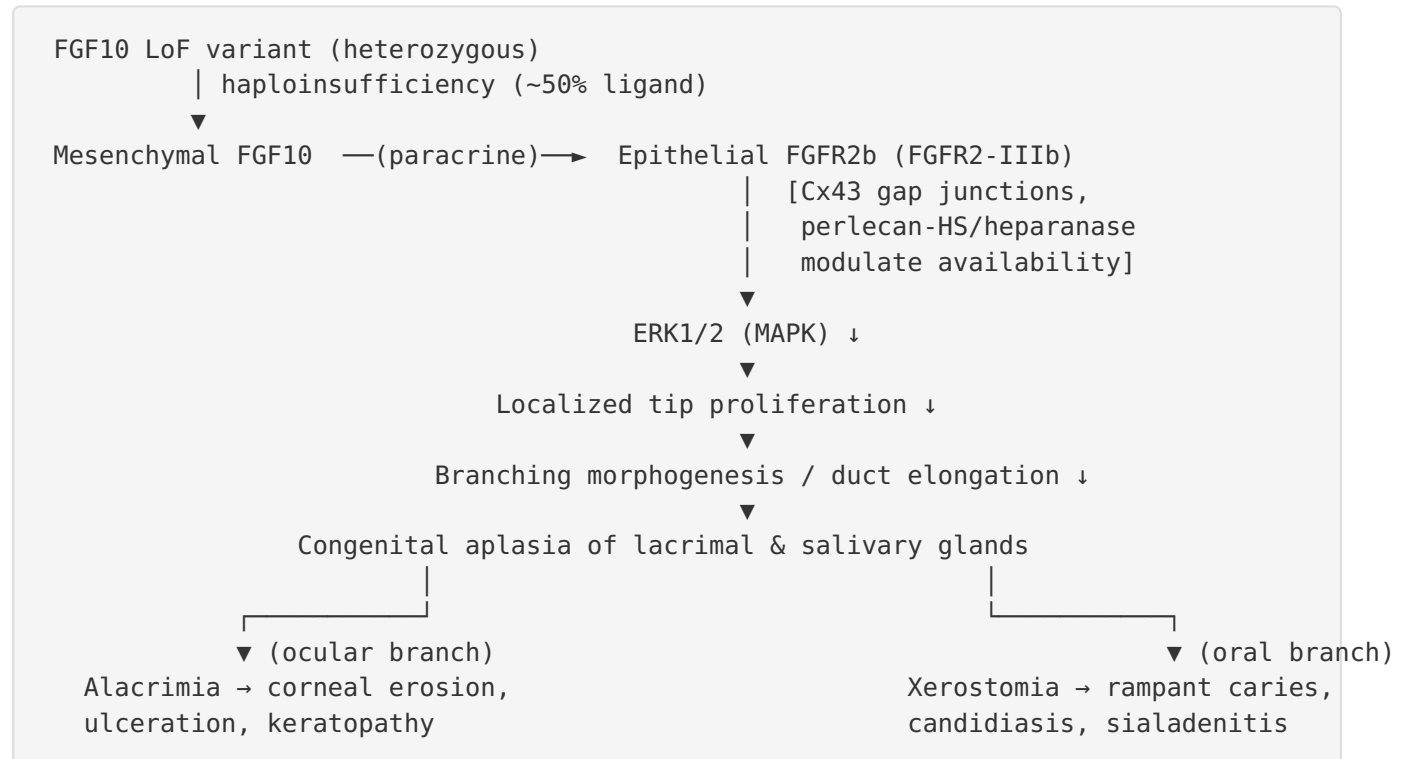
6. Mechanism / Pathophysiology

Ordered causal chain

1. A heterozygous loss-of-function *FGF10* variant (nonsense, splice-site, missense, or whole-gene deletion) **leads to** nonsense-mediated decay or non-functional product, **resulting in** ~50% reduction of functional FGF10 ligand (haploinsufficiency). [PMID: 26955834]
2. Reduced FGF10 secreted by gland **mesenchyme leads to** insufficient activation of epithelial **FGFR2b (FGFR2-IIIb)** receptors on the developing lacrimal and salivary epithelium. [PMID: 10821755]
3. Diminished FGFR2b signaling **results in** reduced downstream **ERK1/2 (MAPK) phosphorylation** and loss of localized epithelial proliferation at duct/bud tips. [PMID: 15716343]
4. Failed proliferation/duct elongation **leads to** defective **branching morphogenesis** of the lacrimal and salivary gland anlagen during embryogenesis. [PMID: 15716343, PMID: 26565022]
5. Defective morphogenesis **results in** congenital **aplasia/atresia/hypoplasia** of lacrimal glands, salivary glands (parotid, submandibular, sublingual), and associated ducts/puncta. [PMID: 10821755]
6. Absent/deficient glands **lead to** reduced tear and saliva production → **xerophthalmia/alacrimia and xerostomia** (and epiphora where drainage structures are atretic). [PMID: 38081329]
7. **Branch A (oral):** Chronic xerostomia **leads to** rampant dental caries (~100%), oral candidiasis, and ascending sialadenitis. [PMID: 22339588, PMID: 11340723]
8. **Branch B (ocular):** Alacrimia **leads to** recurrent corneal erosions, ulceration, scarring, and neovascularization → sight-threatening keratopathy. [PMID: 38537898]

(Steps 1 and 2 are directly demonstrated in human and mouse genetics; steps 3–4 are demonstrated ex vivo in mouse submandibular gland cultures; steps 5–8 are human clinical observations.)

ASCII mechanistic diagram



Supporting detail

- **Molecular pathway (KEGG hsa04010 MAPK; FGF signaling).** FGF10→FGFR2b→RAS/ MAPK(ERK1/2) is the core cascade. *"FGF7 induces epithelial budding, whereas FGF10 induces duct elongation, and both are inhibited by FGFR or ERK1/2 signaling inhibitors"* and *"FGF10 stimulates localized proliferation at the tip of the duct"* (PMID: 15716343).
- **Cellular process.** Epithelial cell proliferation and branching morphogenesis (GO:0048754, branching morphogenesis of an epithelial tube; GO:0060442, branching involved in salivary gland morphogenesis).
- **Co-regulators of ligand availability.** Connexin 43 (Cx43) gap junctions are required for FGF10-induced ERK1/2 phosphorylation and SMG branching (PMID: 26565022); perlecan heparan sulfate and heparanase regulate release and activity of the FGF10–FGFR2b complex in the basement membrane (PMID: 17959718).
- **Extended developmental roles.** FGF10 also governs limb and lung morphogenesis; *Fgf10*-null mice show limb and lung agenesis, rescuable by ubiquitous *Fgf10* (PMID: 23924632), explaining the rare extended ALSG phenotypes (esophageal atresia, pulmonary hypoplasia; PMID: 41409311).

Suggested ontology terms: GO:0060442 (branching involved in salivary gland morphogenesis), GO:0060664 (epithelial cell proliferation involved in salivary gland morphogenesis), GO:0008543 (FGF receptor signaling pathway), GO:0070374 (positive regulation of ERK1/ERK2 cascade). Cell types: CL:0000066 (epithelial cell), CL:0000134 (mesenchymal cell).

7. Anatomical Structures Affected

Organ level (primary). - Lacrimal gland (UBERON:0001817) and lacrimal apparatus/nasolacrimal duct (UBERON:0000395 / UBERON:0004515) - Parotid gland (UBERON:0001831) - Submandibular gland (UBERON:0001736) - Sublingual gland (UBERON:0001833)

Secondary / complication sites. - Cornea (UBERON:0000964) and ocular surface — keratopathy from alacrimia - Teeth (UBERON:0001091) — rampant caries from xerostomia - Oral mucosa — candidiasis

Rare extended involvement (FGF10 pleiotropy). Lung (UBERON:0002048; pulmonary hypoplasia), esophagus (UBERON:0001043; esophageal atresia), and digits/limb (thumb phalangeal hypoplasia), reflecting FGF10's broader developmental roles ([PMID: 41409311](#)).

Body systems: exocrine/glandular (primary), visual/ocular, and digestive (oral) systems.

Tissue and cell level. The affected tissue is **glandular epithelium** (and its inducing **mesenchyme**). The disease is a failure of epithelial branching morphogenesis; FGF10 is expressed in mesenchyme and acts on epithelial FGFR2b.

Subcellular level. FGF10 is a secreted ligand (extracellular/basement membrane, GO:0005615); signaling is transduced at the plasma membrane receptor (FGFR2b) to the cytoplasmic MAPK cascade. No specific organelle pathology (mitochondria, lysosome) is implicated.

Localization / lateralization. Involvement is typically **bilateral** but can be **unilateral or asymmetric**, consistent with variable expressivity — e.g., "*unilateral or bilateral epiphora*" and absence of unilateral lacrimal puncta reported across kindreds ([PMID: 37615310](#), [PMID: 22339588](#)).

8. Temporal Development

Onset. Congenital — the glands fail to form during embryonic development. Symptoms are present from birth (e.g., "*absence of tear secretion since birth*"; PMID: 38537898), though clinical diagnosis often occurs in early childhood (mean ~6.4 years in a salivary agenesis review; PMID: 41791508).

Onset pattern. Chronic/static with respect to the primary structural defect (aplasia does not progress — the glands are simply absent or hypoplastic from the outset).

Progression. The **primary defect is stable** (non-progressive). **Secondary complications are progressive** if untreated: dental caries advance relentlessly under chronic xerostomia, and recurrent corneal erosions lead to cumulative scarring and neovascularization.

Disease course. Chronic, lifelong. Not episodic or relapsing-remitting for the gland defect itself, though complications (caries episodes, corneal erosion episodes, sialadenitis flares) may recur.

Critical periods / windows for intervention. The critical **developmental** window (gland morphogenesis in utero) has passed by birth, so no primary prevention is possible postnatally. The critical **clinical** windows are early childhood for initiating intensive caries prevention and ocular surface protection to prevent irreversible dental and corneal damage.

9. Inheritance and Population

Epidemiology. ALSG is **ultra-rare**. No validated prevalence or incidence figures exist; the disorder is known only from scattered multigenerational kindreds and sporadic case reports. "*Congenital agenesis of the major salivary glands is a rare developmental anomaly*" — a systematic review found only ~20 pediatric cases of complete major salivary gland agenesis (PMID: 41791508). "*Aplasia of lacrimal and salivary glands (ALSG) is a rare autosomal dominant inherited disease*" (PMID: 26955834).

Inheritance pattern. Autosomal dominant.

Penetrance. Incomplete. In a six-generation kindred with a single *FGF10* splice variant (c.429+1G>T), "*one of the variant carriers had no ALSG-related clinical findings, demonstrating incomplete penetrance*" (PMID: 37615310).

Expressivity. Variable, including intrafamilial variability — unilateral vs bilateral involvement, variable dry mouth/carries, and variable lacrimal involvement within the same family carrying one variant (PMID: 37615310).

Recurrence risk. As an autosomal dominant condition, each child of an affected parent has a **50% risk** of inheriting the variant; penetrance and expressivity modify the clinical outcome.

Genetic anticipation, germline mosaicism, founder effects, consanguinity, carrier frequency. No genetic anticipation (not a repeat-expansion disorder). No founder effect, sex predilection, or population enrichment established. "Carrier frequency" is not a standard concept for this dominant, ultra-rare disorder.

Population demographics. No ethnic, geographic, or sex-specific predilection is established; reported kindreds span multiple populations (e.g., Korean LADD family; PMID: 32715658).

10. Diagnostics

Imaging (cornerstone). - **Salivary scintigraphy** with technetium-99m sodium pertechnetate is the principal modality demonstrating bilateral aplasia of parotid, submandibular, and sublingual glands: *"Tc-99m scintigraphy was the most common diagnostic modality"* (PMID: 41791508, PMID: 22339588). - **Orbital CT / cervicofacial MRI** show absent lacrimal, parotid, and submandibular glands, and bony congenital nasolacrimal duct obstruction (PMID: 38081329, PMID: 41533940). - **Ultrasonography** (including UHFUS) can evaluate major and labial salivary/lacrimal glands (PMID: 37685309).

Ocular functional tests. Reduced **Schirmer I test** and shortened **tear breakup time** document tear deficiency: *"Affected patients showed decreased tear production on the Schirmer I test and reduced tear breakup time"* (PMID: 38081329).

Genetic testing. Whole-exome sequencing (WES) identifies causal *FGF10* variants and is the molecular confirmation of choice; single-gene *FGF10* testing and targeted panels (*FGF10/FGFR2/FGFR3* for the ALSG–LADD spectrum) are appropriate. **Chromosomal microarray** detects the 5p12 deletion (PMID: 41533940).

Differential diagnosis. The key differential is **juvenile/primary Sjögren disease** (also causing xerostomia + xerophthalmia), excluded by the congenital onset, absence of autoimmune serology (ANA, anti-SS-A/SS-B), and non-inflammatory imaging/pathology ([PMID: 41791508](#), [PMID: 32715658](#)). Other differentials: LADD syndrome (add ear/dental/digital features), and radiation/iatrogenic gland loss.

Screening. No population newborn screening exists. Cascade genetic testing of at-risk relatives in known kindreds is appropriate given autosomal dominant inheritance.

11. Outcome / Prognosis

Survival / life expectancy. Normal. ALSG is **not life-limiting**; mortality is not increased by the disorder itself.

Morbidity. Substantial but largely preventable. The dominant morbidities are **dental** (rampant caries, potential tooth loss) and **ocular** (keratopathy with risk of vision impairment), plus the quality-of-life burden of chronic dry mouth and dry/irritated eyes.

Complications. Rampant dental caries (~100% of agenesis cases), oral candidiasis, ascending sialadenitis, and recurrent corneal erosions/ulceration with scarring and neovascularization. *"which may cause severe xerostomia, progressive dental caries, and oropharyngeal candidiasis in children"* ([PMID: 22339588](#)); *"repeated episodes of corneal erosions and ulceration and absence of tear secretion since birth"* ([PMID: 38537898](#)).

Recovery potential. The structural gland aplasia is irreversible, but complications are modifiable and partly reversible: minor salivary gland transplantation improved corneal transparency and neovascularization at 5-year follow-up ([PMID: 38537898](#)).

Prognostic factors. Severity of gland involvement (uni- vs bilateral, complete vs partial), timeliness of caries prevention, and ocular surface protection determine long-term outcomes. No molecular prognostic biomarker is established; given variable expressivity, genotype is an imperfect predictor of severity.

12. Treatment

No disease-modifying therapy exists. Management is entirely **symptomatic and supportive**, and because secondary complications are preventable, prevention is the mainstay: *"attention focused on the prevention and treatment of resultant oral disease"* (PMID: 11340723).

Oral / dental management. - Intensive caries prevention: topical/systemic **fluoride**, meticulous oral hygiene, dietary counseling, remineralizing agents. - **Saliva substitutes** and sialogogues for symptomatic relief; restorative dental care for established caries (PMID: 26955834, PMID: 11340723). - Antifungal therapy for oral candidiasis; antibiotics for sialadenitis as needed.

Ocular management. - **Artificial tears / tear substitutes**, lubricating ointments, and ocular surface protection. - **Lacrimal drainage reconstruction** for punctal/duct atresia and epiphora (PMID: 37615310). - **Autologous minor salivary gland transplantation** for severe congenital alacrimia with corneal damage: *"we observed considerable improvement in the photophobia, corneal transparency, and neovascularization"* at long-term (5-year) follow-up (PMID: 38537898).

Advanced / experimental therapeutics. No gene therapy, cell therapy, RNA-based therapy, or targeted pharmacotherapy is in clinical use for ALSG. Given the haploinsufficiency mechanism, ligand-restoration (recombinant FGF10 or dosage-correction) is a conceptual but untested avenue.

Suggested NCIT intervention terms: Fluoride therapy, saliva substitute, artificial tears/ocular lubricant, salivary gland transplantation, lacrimal duct reconstruction, antifungal therapy.

13. Prevention

Primary prevention (of the disease). Not possible postnatally — the glandular defect is established in utero. For families with a known *FGF10* variant, **genetic counseling**, prenatal testing, and preimplantation genetic diagnosis can prevent transmission.

Secondary prevention (early detection). Early clinical recognition and molecular confirmation enable prompt institution of protective measures. Cascade genetic testing identifies at-risk relatives.

Tertiary prevention (of complications) — the practical mainstay. Aggressive, lifelong caries prevention (fluoride, hygiene, diet, regular dental review) and ocular surface protection (lubrication, drainage correction) prevent the major morbidities of ALSG.

Counseling. Autosomal dominant inheritance with 50% recurrence risk, incomplete penetrance, and variable expressivity should be communicated; an unaffected-appearing carrier can still transmit a fully penetrant phenotype to offspring ([PMID: 37615310](#)).

Immunization / public health / environmental prophylaxis. Not applicable (non-infectious, non-environmental disorder).

14. Other Species / Natural Disease

Taxonomy. The disease has been modeled in **mouse** (*Mus musculus*, NCBI:txid10090). No naturally occurring ALSG-equivalent disorder in companion animals or wildlife is documented in the reviewed literature (OMIA entry not identified here).

Orthologous gene. *Fgf10* is highly conserved; mouse *Fgf10* is the functional ortholog used to define the mechanism.

Comparative biology / evolutionary conservation. The FGF10→FGFR2b→ERK1/2 branching-morphogenesis mechanism is deeply conserved across vertebrates, and mouse knockouts faithfully reproduce the human gland-aplasia phenotype (Section 15), indicating strong evolutionary conservation of the disease mechanism.

Transmission / zoonotic potential. Not applicable (genetic developmental disorder).

15. Model Organisms

Primary model — mouse (*Mus musculus*).

- **Knockout:** *Fgf10*-null mice fail to develop lacrimal and salivary glands, directly recapitulating human aplasia. *"Expression of FGF10 in the mesenchyme adjacent to the presumptive lacrimal bud and absence of lacrimal gland development in FGF10-null mice strongly suggest that it is an endogenous inducer"* ([PMID: 10821755](#)). FGF10 is also sufficient to induce ectopic lacrimal bud formation, and *"inhibition of signaling by a receptor for*

FGF10 (receptor 2 IIIb) suppressed development of the endogenous lacrimal bud" (PMID: 10821755), confirming FGFR2-IIIb as the mediating receptor.

- **Phenotype recapitulation:** Excellent for the core defect. Homozygous *Fgf10* null is perinatal-lethal (limb and lung agenesis; PMID: 23924632); the **heterozygous dosage-reduction state** models human haploinsufficiency.
- **Ex vivo organ culture** (embryonic submandibular gland) is a powerful model dissecting FGF10's role in duct elongation/branching via FGFR2b and ERK1/2 (PMID: 15716343), the Cx43 requirement (PMID: 26565022), and heparan-sulfate/heparanase regulation of ligand availability (PMID: 17959718).

Model limitations. Homozygous lethality limits adult phenotyping of complete loss; the mouse does not perfectly replicate the human secondary complications (rampant caries depends on human oral ecology/diet). Variable expressivity and modifier effects seen in human kindreds are not fully captured by inbred knockout lines.

Resources: MGI (*Fgf10*), IMPC/KOMP for allele availability.

Key Findings (Expanded)

F001 — ALSG is an autosomal dominant disorder caused by FGF10 haploinsufficiency. Multiple independent kindreds show heterozygous loss-of-function *FGF10* variants segregating with disease, including a novel nonsense c.237G>A (p.Trp79*) causing NMD-mediated haploinsufficiency (PMID: 26955834) and a donor splice-site c.429+1G>T causing exon 2 skipping (PMID: 37615310). This establishes a clean gene–dosage etiology.

F002 — ALSG and LADD are allelic FGF-signaling disorders. LADD results from heterozygous mutations in *FGFR2*, *FGFR3*, or *FGF10* (PMID: 16501574, PMID: 32715658); ALSG is the restricted, milder phenotype within the same pathway.

F003 / F009 — Mesenchymal FGF10 → epithelial FGFR2b/ERK1/2 drives gland morphogenesis, and mouse knockouts recapitulate the aplasia. FGF10 induces duct elongation via FGFR2b and ERK1/2 and stimulates localized tip proliferation (PMID: 15716343); *Fgf10*-null mice lack lacrimal gland development and blocking FGFR2-IIIb suppresses the endogenous bud (PMID: 10821755). Cx43 and perlecan-HS/heparanase tune ligand availability (PMID: 26565022, PMID: 17959718).

F004 — Incomplete penetrance and variable expressivity. A single splice variant in a six-generation kindred produced unilateral-to-bilateral disease with one non-penetrant obligate carrier (PMID: 37615310); xerostomia 80% and caries 100% in a salivary-agenesis review (PMID: 41791508).

F005–F007 — Diagnosis, complications, and management. Diagnosis uses Tc-99m scintigraphy, orbital/facial imaging, Schirmer test, and genetic testing to exclude Sjögren disease (PMID: 41791508, PMID: 38081329). Complications include rampant caries, candidiasis, sialadenitis, and keratopathy (PMID: 22339588, PMID: 38537898). Management is supportive, with minor salivary gland transplantation benefiting severe alacrimia (PMID: 38537898).

F008 — Extended phenotypic spectrum. WES of bony CNLDO found 8 novel *FGF10* variants plus candidate *IGSF3* variants (PMID: 41533940); a novel LoF variant linked ALSG to esophageal atresia and pulmonary hypoplasia (PMID: 41409311).

Mechanistic Model / Integrated Interpretation

ALSG is best understood as a **dosage-sensitive FGF10 developmental signaling disorder** (Finding F011). The unifying model runs: autosomal dominant *FGF10* haploinsufficiency (PMID: 26955834, PMID: 37615310) → deficient mesenchymal FGF10 → inadequate epithelial FGFR2b/ERK1/2 activation (PMID: 10821755, PMID: 15716343) → failed branching morphogenesis → congenital, variably bilateral aplasia of lacrimal and salivary glands → xerophthalmia/alacrimia and xerostomia → preventable secondary morbidity (rampant caries ~100%, candidiasis, sialadenitis, sight-threatening keratopathy; PMID: 22339588, PMID: 11340723, PMID: 38537898). The disorder is allelic with LADD (PMID: 16501574); the *Fgf10* mouse knockout anchors causality (PMID: 10821755, PMID: 23924632).

ALSG vs LADD comparison

Feature	ALSG	LADD syndrome
Causal genes	<i>FGF10</i> (ligand)	<i>FGF10, FGFR2, FGFR3</i>
Lacrimal involvement	Yes	Yes
Salivary involvement	Yes	Yes
Ear anomalies	No	Yes
Dental anomalies (beyond caries)	No	Yes
Digital anomalies	No	Yes
Inheritance	AD	AD
Relationship	Milder, restricted form	Broader allelic form

Evidence Base

PMID	Role in this report	Source type
26955834	ALSG = AD <i>FGF10</i> haploinsufficiency; nonsense p.Trp79*	Human clinical/genetic
37615310	Incomplete penetrance, variable expressivity; splice variant c.429+1G>T	Human kindred
16501574	LADD caused by FGFR2/FGFR3/FGF10 — allelism	Human genetic
32715658	LADD three-gene model; differential from Sjögren	Human clinical/review
10821755	Mouse <i>Fgf10</i> -null → absent lacrimal gland; FGFR2-IIIb receptor	Model organism
15716343	FGF10→FGFR2b→ERK1/2 drives duct elongation	Ex vivo model
26565022	Cx43 required for FGF10-induced ERK1/2 and branching	Ex vivo model
17959718	Perlecan-HS/heparanase regulate FGF10 availability	Ex vivo model
23924632	<i>Fgf10</i> -null limb/lung agenesis; rescue by ubiquitous Fgf10	Model organism
41791508	Phenotype frequencies; Tc-99m scintigraphy; rarity	Systematic review
22339588	Oral complications; punctal absence	Human case/review
38537898	Minor salivary gland transplant for alacrimia (5-yr)	Human clinical
11340723	Prevention-centered oral management	Human case
38081329	Bony CNLDO phenotype; Schirmer/TBUT	Human clinical
41533940	Variant spectrum; IGSF3 candidate modifier; 5p12 deletion	Human genotype-phenotype
41409311	FGF10 LoF with EA, pulmonary hypoplasia (VACTERL overlap)	Human case
37685309	Ultrasound of exocrine glands (diagnostic context)	Methods

Limitations and Knowledge Gaps

1. **No population epidemiology.** Prevalence/incidence are unknown; the evidence base is limited to ~20 complete-agenesis pediatric cases and scattered kindreds, precluding robust frequency, sex-ratio, or geographic estimates.
 2. **Modifiers unexplained.** Incomplete penetrance and variable expressivity imply genetic/stochastic modifiers that remain uncharacterized; the role of candidate *IGSF3* variants is unresolved ([PMID: 41533940](#)).
 3. **Genotype–phenotype correlation is weak.** The same variant produces unilateral to bilateral disease within one family, so genotype poorly predicts severity.
 4. **No disease-modifying therapy or trials.** No ClinicalTrials.gov interventions target the FGF10 axis in ALSG; ligand-restoration is untested.
 5. **Human molecular profiling absent.** No transcriptomic/proteomic/metabolomic patient datasets exist; mechanism is inferred from mouse and ex vivo systems.
 6. **No natural animal disease / OMIA entry identified,** limiting comparative pathology.
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Proposed Follow-up Experiments / Actions

1. **Establish an international ALSG/LADD registry** to aggregate genotype, phenotype laterality, penetrance, and natural-history data, enabling the first prevalence and genotype–phenotype estimates.
2. **Test *FGF10* modifier hypotheses** — sequence *IGSF3* and perform exome-wide modifier screens in penetrant vs non-penetrant carriers within multigenerational kindreds.
3. **Generate heterozygous *Fgf10* dosage mouse models** (and conditional epithelium/mesenchyme alleles) to quantify the gland-formation dosage threshold and phenotype laterality, directly modeling human haploinsufficiency.
4. **Patient-derived organoid/iPSC salivary and lacrimal gland models** to functionally classify missense VUS and screen FGF10-pathway agonists (recombinant FGF10, FGFR2b activators, heparanase modulators).
5. **Prospective evaluation of minor salivary gland transplantation** for alacrimia across more patients with standardized corneal and quality-of-life endpoints, building on the encouraging 5-year single-series result.

6. Standardize a diagnostic algorithm (scintigraphy + orbital/cervicofacial imaging + Schirmer/TBUT + *FGF10/FGFR2/FGFR3* sequencing) with explicit criteria to distinguish congenital ALSG from juvenile Sjögren disease.

Report compiled from 11 confirmed findings and 21 reviewed papers over a 5-iteration autonomous investigation. Evidence source types are indicated throughout as human clinical/genetic, model organism, or ex vivo.

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